In the circuit given below:

Load 1 absorbs a **complex** power of **150 VA** at a **leading** power factor of 0.4

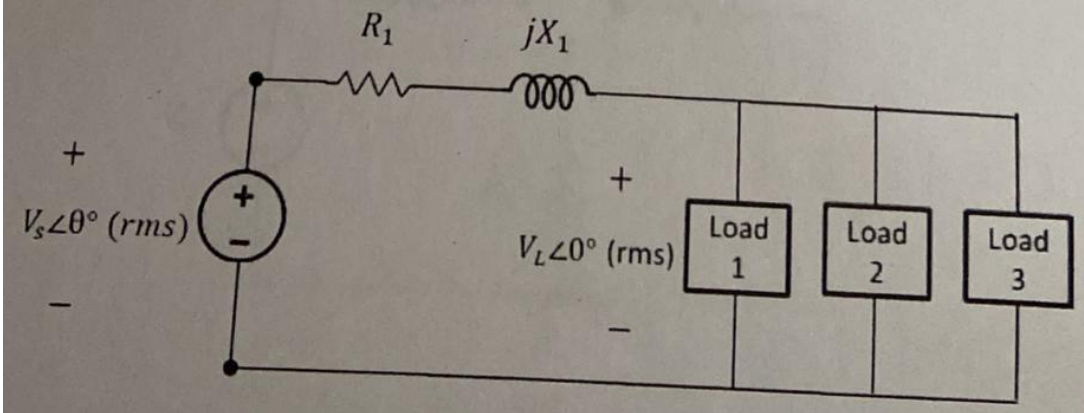
Load 2 absorbs an **active** power of **680 W** at a **lagging** power factor of 0.6

Load 3 absorbs a **reactive** power of **390 VAr** at a **lagging** power factor of 0.6

(15 pts.) 3.a) Find V_s (rms) and θ for the **voltage source**.

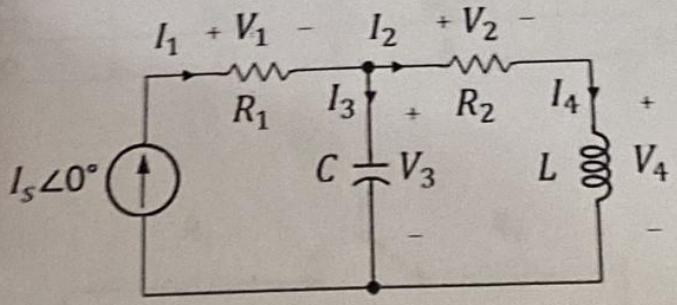
(10 pts.) 3.b) Construct the **phasor diagram** showing all the voltage and current phasors.

R_1 (Ω)	jX_1 (Ω)	V_L (V)
8	14	120



The circuit given below has a **sinusoidal current** source with $45\angle 0°$ A rms **current** and $\omega = 1500\ rad/s$, one resistor with 240 Ω, an inductor with 80 mH and a capacitor with 0.9 μF. Please answer the following:

(30 pts.) For each passive element in the circuit, find the rms phasor voltages (V) and currents (I) together with the real powers (P) and magnetizing reactive powers (Q). Please **write each step of your calculations in your solution clearly**. You can write your final results in a table similar to the one given below.

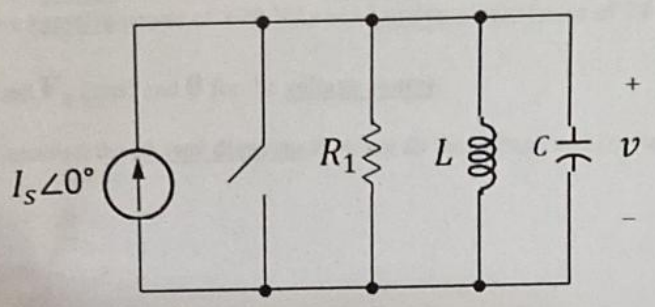


An **example table format** to provide your results, **after** presenting your calculations.

	Resistor 1	Resistor 2	Capacitor	Inductor
V (Volts rms)				
I (Ampers rms)				
P (Watts)				
Q (VAr)				

Two circuits are given below:

(15 pts.) 4. **No energy** stored in the 40 mH inductor or the 30 nF capacitor, when the switch in the circuit given below is closed. The resistor in the circuit is 500 Ω. The current source is DC and its value is 28 mA. Find $i_L(t)$ for $t \geq 0$.



Formula's that can be used in this question (**depending on damping**):

$$\alpha = \frac{1}{2RC}\ rad/s$$

$$\omega_0 = \frac{1}{\sqrt{LC}}\ rad/s$$

$$\omega_d = \sqrt{{\omega_0}^2 - \alpha^2}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - {\omega_0}^2}$$

$$s_2 = -\alpha - \sqrt{\alpha^2 - {\omega_0}^2}$$

Damping	Step Response Equations	Coefficient Equations
Overdamped	$x(t) = X_f + A_1' e^{s_1 t} + A_2' e^{s_2 t}$	$x(0) = X_f + A_1' + A_2'$ $dx/dt\,(0) = A_1' s_1 + A_2' s_2$
Underdamped	$x(t) = X_f + (B_1' cos\omega_d t + B_2' sin\omega_d t) e^{-\alpha t}$	$x(0) = X_f + B_1'$ $dx/dt\,(0) = -\alpha B_1' + \omega_d B_2'$
Critically damped	$x(t) = X_f + D_1' t e^{-\alpha t} + D_2' t e^{-\alpha t}$	$x(0) = X_f + D_2'$ $dx/dt\,(0) = D_1' - \alpha D_2'$

The uncharged capacitor in the given circuit is initially connected to terminal d of the three-position switch. At $t = 0$, the switch is moved to position e, where it remains for **4 ms**. After that, the switch is moved to position f where it remains indefinitely. Please answer the following:

(20 pts.) 2.a) Using **the general solution method**, derive the numerical expressions for the voltage across the capacitor when the switch was moved **to terminals *e* and *f*** (one expression **for each switch position**).

(10 pts.) 2.b) When will the voltage on the capacitor equal $\frac{V_s}{6}$? (if more than one time, please provide each).

R_1 (kΩ)	R_2 (kΩ)	R_3 (kΩ)	C (μF)
20	12	8	0.4

